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Hybrid flow-shop manufacturing network reliability optimization using genetic algorithm and absorbing Markov chain

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ABSTRACT

In this study, we adopt a strategy of machine configuration to construct a reliable hybrid flow-shop manufacturing system (HFSMS) with reworking and scrapping actions, in which the reworking and scrapping actions are due to the yield rates of the configured machines. The machine configuration determines the machine suppliers and the number of machines for each production stage in the HFSMS, which may affect the stability of the HFSMS. The machines configured to a production stage in parallel are provided by the same supplier. Accordingly, each production stage has multiple states, following a probability distribution. The HFSMS is a typical stochastic-flow network under machine configuration, and network reliability indicates the probability that order demand d can be fulfilled by the HFSMS and used as a decision reference. Our study integrates genetic algorithm (GA) and absorbing Markov chain (AMC) to solve the reliability-oriented machine configuration problem of HFSMS subject to a configuration budget, where the GA is utilized to find the optimal machine configuration with maximum network reliability and the AMC model constructed based on the yield rates is used for network reliability evaluation. For validating the applicability and computational efficiency of the proposed AMC and GA-based approach, a simple manufacturing system and two practical manufacturing systems are used to compare the proposed approach with four popular meta-heuristic algorithms. The experimental results show that the proposed approach can find the exact optimal solution and has better computational efficiency than the other four meta-heuristic algorithms.

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1. Introduction

A hybrid flow-shop manufacturing system (HFSMS) typically contains multiple production lines to manufacture the same product for order demand sharing [1,2]. Each product is processed through a production line, consisting of a series of production stages to become a finished good [3,4]. Additionally, each production stage may have only one or multiple machines in parallel to provide capacity [5,6]. Because of the machine's yield rate, each production stage may produce defective prod-

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ucts, increasing the manufacturing cost [7]. Hadjinicola [8] and Sarker, Jamal, and Mondal [9] suggested that correcting poorly manufactured products can benefit manufacturers. Therefore, Al-Salamah [10], Chiu, Chen, Wu, and Chiu [11], and Colledani and Angius [12] suggested reworking mechanisms to decrease the number of scraped products in manufacturing systems and evaluated the delivery time or predicted the throughput. However, their proposed model is deterministic, indicating that the production capacity of the manufacturing systems is not stochastic.

From the maintenance perspective, Liu et al. [13] considered machine failures and focused on integrating production planning and preventive maintenance scheduling for advanced manufacturing systems with synchronized parallel machines. Liu et al. [14] devoted to determining optimal selective maintenance and inspection strategy based on the distributions of components' states such that the probability of the system completing the next mission can be significantly increased. Li et al. [15] proposed a predictive maintenance strategy to minimize maintenance costs and economic losses based on two reliability metrics for manufacturing systems driven by product defects and machine degradation. These studies have one thing in common: machine/component failure or maintenance may occur with a probability distribution. In other words, the capacity of a production stage with parallel machines may have multiple statuses in reality. Thus the HFSMS can be viewed as a stochastic-flow network [16,17] with nodes representing production stages and arcs representing conveyors. Therefore, network reliability is defined as the probability that the HFSMS can meet an order demand d and is an important performance index [18–22], in which Lin and Huang [18] evaluated the network reliability with a due date, but the reworking characteristic is not taken into account. For calculating the exact network reliability for the HFSMS with reworking actions, all the minimal capacity vectors for order demand d (called d-MCVs) according to the flow of product at each production stage are needed and are generally determined by a graphic technique [19–21]. For instance, Zhang et al. [23] used the graphic technique to evaluate the network reliability of a seru manufacturing system with reworking actions. Nevertheless, the graphic technique for obtaining all d-MCVs required enumerating all regular and reworking paths, which is time-consuming for more complex HFSMS.

To determine all d-MCVs, an absorbing Markov chain (AMC) is used. The AMC is a type of Markov Chain. The probability of each state in the AMC model can only be determined using the previous state, representing random transitions between a pair of states, and there exists at least one absorbing state [24]. The product flow traveling through a manufacturing network can be viewed as a state transition process. Therefore, several studies have used the AMC to calculate the reworking proportion or the required raw material for the manufacturing systems, in which two absorbing states represent the product flow output and scrap [25–27]. Alternatively, the AMC can determine the product flow passing through each production stage. Hence, the AMC can enumerate all d-MCVs in the HFSMS.

Furthermore, constructing a reliable HFSMS is critical despite the global economy, particularly for the HFSMS design phase. Designing an HFSMS involves machine configuration, which includes determining machine suppliers and numbers for each production stage [28,29]. Saxena and Jain [30] indicated that the machine configuration directly impacts overall system performance and that optimizing the machine configuration for a manufacturing system is critical. Niroomand [31] utilized simulated annealing (SA) to configure a set of equipment to stations of an assembly line for station establishment cost and equipment purchasing cost minimization. Sabioni, Daaboul, and Duigou [32] considered the machine layout and the product configuration driven by customer requirements to determine the optimal reconfigurable manufacturing system with minimum overall manufacturing cost. They represented the problem as a $0-1$ nonlinear integer programming model and combined genetic algorithm (GA) and a brute-force algorithm to solve it. Chang and Huang [33] concurrently determine the optimal machine types and buffer size allocation for production line capacity maximization. They showed that the proposed machine selection and buffer allocation algorithm outperforms existing approaches, including GA and tabu search.

Although these studies involve the concept of machine configuration, they do not consider stochastic production capacity. Therefore, our study focuses on searching for the optimal machine configuration of HFSMS with maximum network reliability subject to a configuration budget, in which a machine has two statuses: fail or provide a production capacity, and the configuration cost and the product flow may be reworked or scraped at the production stage. Jahromi and Tavakkoli-Moghaddam [34] demonstrated that determining the optimal machine configuration is an NP-hard problem. Meta-heuristic algorithms, such as GA, particle swarm optimization (PSO), and SA, are suitable for this problem. Additionally, Lin and Yeh [35], Yeh, Fiondella and Chang [36], and Yeh [37] have demonstrated that GA can efficiently solve stochastic-flow network reliability optimization problems. An approach integrating GA and AMC, known as AMC-GA, is proposed to address this problem. The main contributions of this paper are as follows:

(1) From the perspective of HFSMS system design, a reliability-oriented machine configuration problem is discussed. Its mathematical model is defined.
(2) A stochastic-flow network of the HFSMS associated with a machine configuration is modeled. And then, an algorithm based on AMC to calculate network reliability is developed, and its computational efficiency is shown by comparing it with a graphic technique.
(3) The AMC-GA is proposed to solve the addressed problem. A simple HFSMS and two practical HFSMSs are adopted to demonstrate its applicability and computational efficiency by comparing it with several popular meta-heuristic algorithms.

The remainder of this study is organized as follows. Section 2 defines the mathematical model of the addressed problem. Subsequently, an AMC-based algorithm is proposed to evaluate the HFSMS network reliability in Section 3. Section 4 illustrates how to use the AMC-GA to search for the optimal machine configuration. Several numerical experiments are executed

to demonstrate the applicability and computational efficiency of the AMC-GA in Section 5. Finally, conclusions and future research are provided in Section 6.

2. Mathematical model for HFSMS machine configuration of reliability optimization

The HFSMS is constructed with m parallel identical production lines, L_1, L_2, ..., L_m, and each production line contains n production stages. Let (**E**, **V**) be the HFSMS with **E** representing the set of perfectly reliable arcs (conveyors) and $\mathbf{V} = \{v_{j,i} | j = 1, 2, ..., m, i = 1, 2, ..., n\}$ representing the set of vertices (production stages). The symbol $v_{j,i}$ represents the ith production stage of the jth production line. The raw material input and the product output of the production line L_j are represented by I_j and O_j, respectively, for $j = 1, 2, ..., m$. In each production line, there are z reworking stages. When a defect in the product flow is detected at the reworking stage v_{j,β_e}, it will return to the indicated stage v_{j,α_e} to be reworked for $e = 1, 2, ..., z$ and $j = 1, 2, ..., m$, where v_{j,β_e} with $\beta_e \in \{1, 2, ..., n\}$ represents the eth reworking stage in the jth production line and v_{j,α_e} with $\alpha_e \in \{1, 2, ..., \beta_e\}$ is the starting stage of reworking stage v_{j,β_e}. For instance, the following figure shows that the HFSMS (**E**, **V**) includes two production lines, each with two reworking stages.

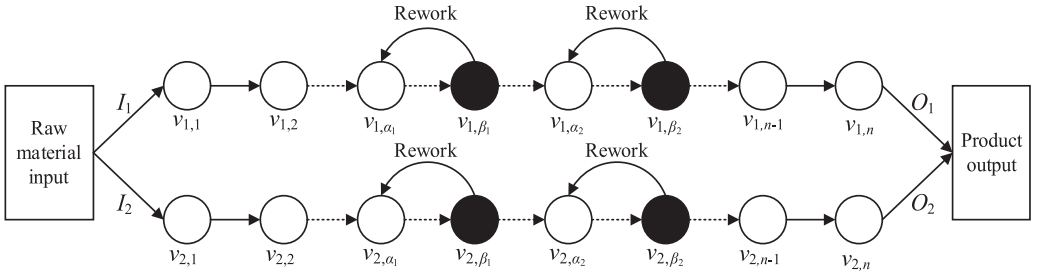


Fig. 1. A reworkable HFSMS with two production lines.

For the ith production stage in the HFSMS (**E**, **V**), there are s_i suppliers providing machines. A machine for the ith production stage (called stage-i machine for short), provided by an uth supplier, has the following attributes: unit cost $c_{u,i}$, machine reliability $r_{u,i}$, production capacity $k_{u,i}$, and yield rate $p_{u,i}$ for $u = 1, 2, ..., s_i$ and $i = 1, 2, ..., n$. Assume that M_i^u represents the available stage-i machine quantity of the uth supplier. Let (T, Y) be a machine configuration where $T = (t_{1,1}, t_{2,1}, ..., t_{1,n}, t_{2,1}, t_{2,2}, ..., t_{2,n}, ..., t_{m,1}, t_{m,2}, ..., t_{m,n})$ represents a supplier selection vector and $Y = (y_{1,1}, y_{2,1}, ..., y_{1,n}, y_{2,1}, y_{2,2}, ..., y_{2,n}, ..., y_{m,1}, y_{m,2}, ..., y_{m,n})$ represents a machine quantity vector. Both decision variables $t_{j,i} \in \{1, 2, ..., s_i\}$ and $y_{j,i} \in \{1, 2, ..., M_i^{t_{j,i}}\}$, represent the selected supplier and the number of identical machines configured in parallel for the production stage $v_{j,i}$, respectively.

The current production status of (**E**, **V**) under a machine configuration (T, Y) is represented as $X = (x_{1,1}, x_{2,1}, ..., x_{1,n}, x_{2,1}, x_{2,2}, ..., x_{2,n}, ..., x_{m,1}, x_{m,2}, ..., x_{m,n})$, where $x_{j,i}$ is the current capacity of production stage $v_{j,i}$. Because the production stage $v_{j,i}$ is composed of $y_{j,i}$ machines provided by supplier $t_{j,i}$ and each machine operating with capacity $k_{t_{j,i},i}$ and probability $r_{t_{j,i},i}$, each $x_{j,i}$ may take one of the following values: 0, $k_{t_{j,i},i}$, $2k_{t_{j,i},i}$, ..., $y_{j,i}k_{t_{j,i},i}$, and follows the Binomial distribution:

$$\Pr(x_{j,i} = \tau k_{t_{j,i},i}) = \binom{y_{j,i}}{\tau} (r_{t_{j,i},i})^{\tau} (1 - r_{t_{j,i},i})^{y_{j,i}-\tau}, \tag{1}$$

where $\tau \in \{0, 1, 2, ..., y_{j,i}\}$ means the number of normal machines in $v_{j,i}$.

Assume that $\mathbf{X}_{(T,\ Y),d}$ represents the set of all production statuses of (**E**, **V**) fulfilling the order demand d under machine configuration (T, Y). Therefore, $R_d(T, Y)$ represents the network reliability associated with machine configuration (T, Y) as follows:

$$R_d(T, Y) = \sum\nolimits_{X \in \mathbf{X}_{(T,Y),d}} \Pr(X), \tag{2}$$

where $\Pr(X) = \Pr(x_{1,1}) \times \Pr(x_{1,2}) \times ... \times \Pr(x_{m,n})$, and $\Pr(x_{j,i})$ is driven by Eq. (1) for each j and i. Furthermore, the total cost of machine configuration (T, Y), represented by $C(T, Y)$, is expressed as follows:

$$C(T, Y) = \sum\nolimits_{j=1}^{m} \sum\nolimits_{i=1}^{n} c_{t_{j,i},i} y_{j,i}, \tag{3}$$

where $c_{t_{j,i},i} y_{j,i}$ shows the configuration cost of $v_{j,i}$. Given the configuration budget B, the mathematical model for the discussed problem is given as follows:

$$\textit{Maximize } R_d(T, Y) = \sum_{X \in \mathbf{X}_{(T,Y),d}} \Pr(X), \tag{4}$$

Subject to

$$C(T, Y) \le B, \tag{5}$$

$$\sum_{j:t_{j,i}=u} y_{j,i} \le M_i^u \text{ for } u = 1, 2, \ldots, s_i \text{ and } i = 1, 2, \ldots, n, \tag{6}$$

$$t_{j,i} \in \{1, 2, \ldots, s_i\} \text{ for } i = 1, 2, \ldots, n \text{ and } j = 1, 2, \ldots, m, \text{ and} \tag{7}$$

$$y_{j,i} \in \left\{1, 2, \ldots, M_i^{t_{j,i}}\right\} \text{ for } i = 1, 2, \ldots, n \text{ and } j = 1, 2, \ldots, m. \tag{8}$$

Constraint (5) indicates that the cost of the machine configuration cannot exceed budget B. Constraint (6) indicates that the number of machines a supplier provides cannot exceed its maximal supply. Constraints (7) and (8) describe the scope of both decision variables $t_{j,i}$ and $y_{j,i}$, respectively. The optimal solution is obtained by maximizing the objective function (4). To solve the optimization problem, the following assumption must be made:

(1) To ensure product consistency, the machines configured to the same stage $v_{j,i}$ must be from the same supplier.
(2) Each production stage's status must be statistically independent even if the HFSMS is supported by multiple machine suppliers.
(3) Each product flow traveling through a reworking stage must be reworked at most once. If the product flow fails after reworking, it will be scrapped.
(4) Because production lines may be located at different places, the product flow traveling through a production line can not be transferred to others.

3. AMC for HFSMS network reliability evaluation under a machine configuration

Computing network reliability $R_d(T, Y) = \sum_{X \in \mathbf{X}_{(T,Y),d}} \Pr(X)$ by summing up the probabilities of all $X \in \mathbf{X}_{(T,Y),d}$ may be time-consuming. Lin and Chang [19,20] suggested searching for all d-MCVs and then representing the network reliability as a probability with a union form of the d-MCVs such that the probability can be calculated using a Recursive Sum of Disjoint Products (RSDP) [38], Inclusion-Exclusion Principle [39], or State Space Decomposition [40]. Therefore, this section first introduces the AMC model for d-MCV enumeration. Subsequently, an AMC-based algorithm is presented to calculate the network reliability through the d-MCVs. Finally, the AMC-based algorithm is compared with the graphic technique, which is reference with Lin and Chang [19,20], to demonstrate its computational efficiency.

3.1. AMC model for d-MCV enumeration

The product flow processed in a production line can be illustrated in the AMC model. Let $\Psi_e^j = \{v_{j,\alpha_e}, v_{j,\alpha_e+1}, \ldots, v_{j,\beta_e-1}, v_{j,\beta_e}\}$ be an ordered node set or process in which the defective flow at v_{j,β_e} return to be reworked from v_{j,α_e} to v_{j,β_e} for $e = 1, 2, \ldots, z$ and $j = 1, 2, \ldots, m$. The AMC model of the production line L_j has $n + \sum_{e=1}^{z} |\Psi_e^j| + 2$ states, in which the first n transient states describe the regular flow processed at the production stages, the following $\sum_{e=1}^{z} |\Psi_e^j|$ transient states describe the reworked flow processed at the corresponding production stages, and the last two absorbing states, represent the scraped flow and the finished product output. The AMC transition probability matrix $\mathbf{P}^j$ of production line L_j is thus constructed in the following canonical form.

$$\mathbf{P}^j = \begin{bmatrix} \mathbf{Q}^j & \mathbf{R}^j \\ \mathbf{0} & \mathbf{I} \end{bmatrix}. \tag{9}$$

The term $\mathbf{Q}^j$ is a $n + \sum_{e=1}^{z} |\Psi_e^j|$ by $n + \sum_{e=1}^{z} |\Psi_e^j|$ matrix, in which each element includes the transition probability of an ordered pair of two transient states. The term $\mathbf{R}^j$ is a $n + \sum_{e=1}^{z} |\Psi_e^j|$ by 2 matrix, in which each element contains the transition probability from a transient state to an absorbing state. The term $\mathbf{0}$ represents a 2 by $n + \sum_{e=1}^{z} |\Psi_e^j|$ zero matrix, and the term $\mathbf{I}$ represents a 2 by 2 identity matrix. All transition probabilities in $\mathbf{Q}^j$ and $\mathbf{R}^j$ are given based on the yield rates of the configured machines.

The AMC transition probability matrix can also be described using a transition diagram. For example, Fig. 2 is a transition diagram that shows the AMC transition probability matrix of the production line L_j in Fig. 1, for $j = 1$ and 2. Each frame in Fig. 2 stands for a state. The process from state 1 to state n means the regular process. The states, no. $n + 1$ to no. $n + |\Psi_1^j| + |\Psi_2^j|$ constitute the reworking processes, and both absorbing states, no. $n + |\Psi_1^j| + |\Psi_2^j| + 1$ and no. $n + |\Psi_1^j| + |\Psi_2^j| + 2$, represent the scrapping and successful output states, respectively. Note that some states appear in the regular process, and the reworking process represents the same production stage. For example, both states α_1 and $n + 1$ belong to the production stage v_{j,α_1} .

Because the HFSMS ($\mathbf{E}$, $\mathbf{V}$) owns m production lines, these production lines can share the production of the order demand d such that

$$d = \sum_{j=1}^{m} d_j, \tag{10}$$

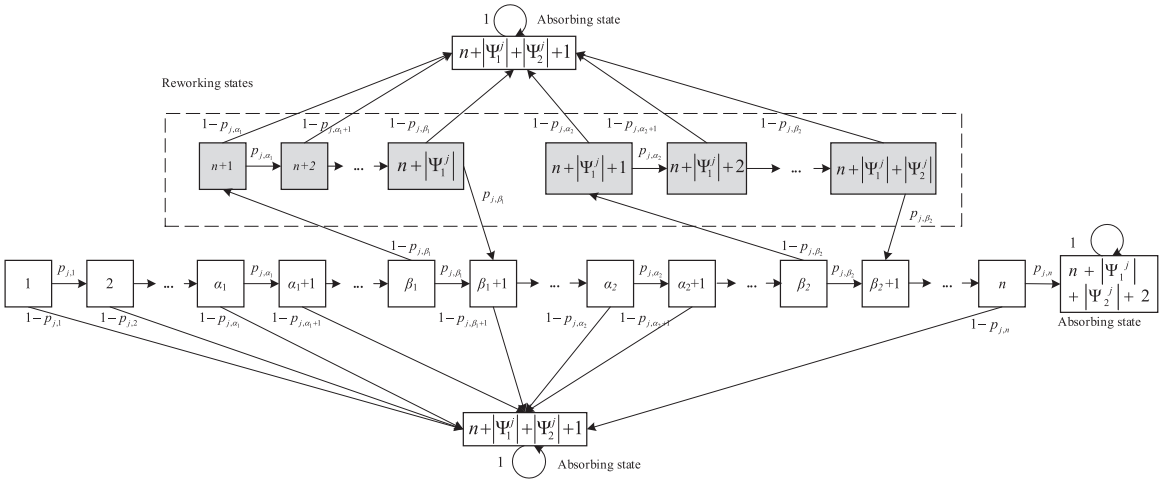


Fig. 2. Transition diagram of production line L_j in Fig. 1.

where d_j is the amount assigned to the jth production line to manufacture. Suppose $D = (d_1, d_2, ..., d_m)$ is a demand pattern fulfilling Constraint (10). Let $I = (I_1, I_2, ..., I_m)$ be a raw material vector and $G = (g_{1,1}, g_{1,2}, ..., g_{1,n}, g_{2,1}, g_{2,2}, ..., g_{m,n})$ be a flow vector, where $g_{j,i}$ is the product flow entering production stage $v_{j,i}$). To find a d-MCV that satisfies the demand pattern D, it is required to compute the required raw material vector I and the corresponding flow vector G.

Let $\mathbf{\Lambda}_{j,i}$ be a set of the states belonging to the same production stage $v_{j,i}$ in the transition diagram. For instance, in Fig. 2, $\mathbf{\Lambda}_{j,2} = \{2\}$, $\mathbf{\Lambda}_{j,\alpha_1} = \{\alpha_1, n+1\}$, and $\mathbf{\Lambda}_{j,\alpha_1+1} = \{\alpha_1+1, n+2\}$. Eq. (11), which is proposed by White [24], can be used to calculate the expected number of times in (transient) state μ with the given starting state ν.

$$\mathbf{E}^j = [\mathbf{I} - \mathbf{Q}^j]^{-1}, \tag{11}$$

in which E^j represents the expected time matrix of the production line L_j. The element in the first row and the μth column of $\mathbf{E}^j$, denoted by $E^j_{1,\mu}$, is the expected times of a unit product arrives at a state μ from state 1 before arriving at the absorbing state. Eq. (12), which is proposed by White [24], is subsequently used to calculate the absorbing probability at the absorbing state μ with a given starting state ν.

$$\mathbf{A}^j = \mathbf{E}^j\mathbf{R}^j, \tag{12}$$

where $\mathbf{A}^j$ is the absorbing probability matrix of the production line L_j. The first row in $\mathbf{A}^j$ includes two elements of the scrapping rate $A^j_{1,1}$ and the successful output rate $A^j_{1,2}$, where $A^j_{1,1}$ (resp. $A^j_{1,2}$) indicates the probability that a unit of raw material input processed by production line L_j is scraped (resp. successful). Accordingly, the required raw material input I_j of the production line L_j to produce d_j can be determined using Eq. (13):

$$I_j = d_j / A^j_{1,2} \text{ for } j = 1, 2, \ldots, m. \tag{13}$$

The corresponding product flow $g_{j,i}$ passing through production stage $v_{j,i}$ can thus be determined using the following equation:

$$g_{j,i} = I_j \sum_{\mu \in \mathbf{\Lambda}_{j,i}} E^j_{1,\mu} \text{ for } i = 1, 2, \ldots, n \text{ and } j = 1, 2, \ldots, m. \tag{14}$$

The flow vector G generated by the AMC is feasible under machine configuration (T, Y) if it satisfies the following constraint:

$$g_{j,i} \le y_{j,i} k_{t_{j,i},i} \text{ for } i = 1, 2, \ldots, n \text{ and } j = 1, 2, \ldots, m. \tag{15}$$

That is, under machine configuration (T, Y), each product flow $g_{j,i}$ cannot exceed the maximal production capacity of production stage $v_{j,i}$. For convenience, let $\mathbf{G}_{(T,\ Y)}$ be the set of the flow vectors generated by the AMC that satisfy Constraint (15). Lin and Chang [19] defined a d-MCV as any production capacity vector X transformed from a flow vector $G \in \mathbf{G}_{(T,\ Y)}$ using the following equation:

$$x_{j,i} = \theta k_{t_{j,i},i} \text{ if } (\theta - 1) k_{t_{j,i},i} < g_{j,i} \le \theta k_{t_{j,i},i}, \text{ where } \theta \in \{0, 1, 2, \ldots, y_{j,i}\} \text{ for } i = 1, 2, \ldots, n \text{ and } j = 1, 2, \ldots, m. \tag{16}$$

3.2. Development of an AMC-based algorithm

Let $\mathbf{\Omega}_{(T,\ Y)} = \{X^1, X^2, ..., X^\omega\}$ be a set of d-MCVs. Therefore, network reliability under (T, Y) is represented as the probability of the union of the d-MCVs, as follows:

$$R_d(T,Y) = \Pr\{\cup_{h:X^h\in\mathbf{\Omega}_{(T,Y)}}\{X|X \geq X^h\}\}, \tag{17}$$

where $\Pr\{X|X \geq X^h\} = \Pr(x \geq x^h_{1,1}) \times \Pr(x \geq x^h_{1,2}) \times \ldots \times \Pr(x \geq x^h_{m,n})$. Recently, many studies have used the RSDP to calculate the probability of the union form because of its efficient computation capability [35,36,41]. Hence, the RSDP is used in this study.

Based on the above proposed AMC model, an AMC-based algorithm integrating the RSDP is developed to calculate network reliability as follows:

AMC-based Algorithm

Input: Demand d, HFSMS $(\mathbf{E}, \mathbf{V})$ including production lines $\{L_1, L_2, ..., L_m\}$, reworking stages $\{v_{j,\beta_1}, v_{j,\beta_2}, ..., v_{j,\beta_z}\}$ and starting stages $\{v_{j,\alpha_1}, v_{j,\alpha_2}, ..., v_{j,\alpha_z}\}$, and machine configuration (T, Y), each configured machine's production capacity $k_{t_{j,i},i}$, reliability $r_{t_{j,i},i}$, and yield rate $p_{t_{j,i},i}$.

Output: $R_d(T, Y)$.

Step 1. Calculate the probability distribution of each production stage under machine configuration (T, Y).

$$\Pr\left(x_{j,i} = \tau k_{t_{j,i},i}\right) = \binom{y_{j,i}}{\tau}\left(r_{t_{j,i},i}\right)^{\tau}\left(1 - r_{t_{j,i},i}\right)^{y_{j,i}-\tau} \text{ for } t = 0, 1, 2, \ldots, y_{j,i},\ i = 1, 2, \ldots, n, \text{ and } j = 1, 2, \ldots, m. \tag{18}$$

Step 2. Find all demand pattern $D = (d_1, d_2, ..., d_m)$ satisfying the following constraint:

$$d = \sum_{j=1}^{m} d_j. \tag{19}$$

Step 3. For each production line L_j, repeat the following steps.

3.1 Construct the AMC transition probability matrix $\mathbf{P}^j$ of production line L_j.

$$\mathbf{P}^j = \begin{bmatrix} \mathbf{Q}^j & \mathbf{R}^j \\ \mathbf{0} & \mathbf{I} \end{bmatrix}. \tag{20}$$

3.2 Generate the expected time matrix $\mathbf{E}^j$ of production line L_j.

$$\mathbf{E}^j = [\mathbf{I} - \mathbf{Q}^j]^{-1}. \tag{21}$$

3.3 Find the successful output rate $A^j_{1,2}$. ($E^j_{1,*}$ is the first row of matrix $\mathbf{E}^j$ and $R^j_{*,2}$ is the last column of matrix $\mathbf{R}^j$)

$$A^j_{1,2} = E^j_{1,*}R^j_{*,2}. \tag{22}$$

Step 4. Repeat the following steps for each feasible demand pattern D from Step 2.

4.1 Determine the required raw material input I_j for each production line via the following equation.

$$I_j = d_j/A^j_{1,2} \text{ for } j = 1, 2, \ldots, m. \tag{23}$$

4.2 Compute the product flow $g_{j,i}$ passing through production stage $v_{j,i}$.

$$g_{j,i} = I_j \sum_{\mu\in\mathbf{\Lambda}_{j,i}} E^j_{1,\mu} \text{ for } i = 1, 2, \ldots, n \text{ and } j = 1, 2, \ldots, m. \tag{24}$$

4.3 Confirm whether the flow vector G from Step 4.2 satisfies the following constraint.

$$g_{j,i} \leq y_{j,i}k_{t_{j,i},i} \text{ for } i = 1, 2, \ldots, n \text{ and } j = 1, 2, \ldots, m. \tag{25}$$

4.4 Transform each feasible flow vector G from Step 4.3 into a d-MCV X via the following equation.

$$x_{j,i} = \theta k_{t_{j,i},i} \text{ if } (\theta - 1)k_{t_{j,i},i} < g_{j,i} \leq \theta k_{t_{j,i},i}, \text{ where } \theta \in \left\{0, 1, 2, \ldots, y_{j,i}\right\} \text{ for } i = 1, 2, \ldots, n \text{ and } j = 1, 2, \ldots, m. \tag{26}$$

Step 5. Suppose $\mathbf{\Omega}_{(T,\ Y)} = \{X^1, X^2, ..., X^\omega\}$ is the set of all d-MCVs obtained from Step 4. Apply the RSDP to calculate the network reliability $R_d(T, Y) = \Pr\{\cup_{h:X^h\in\mathbf{\Omega}_{(T,Y)}}\{X|X \geq X^h\}\}$.

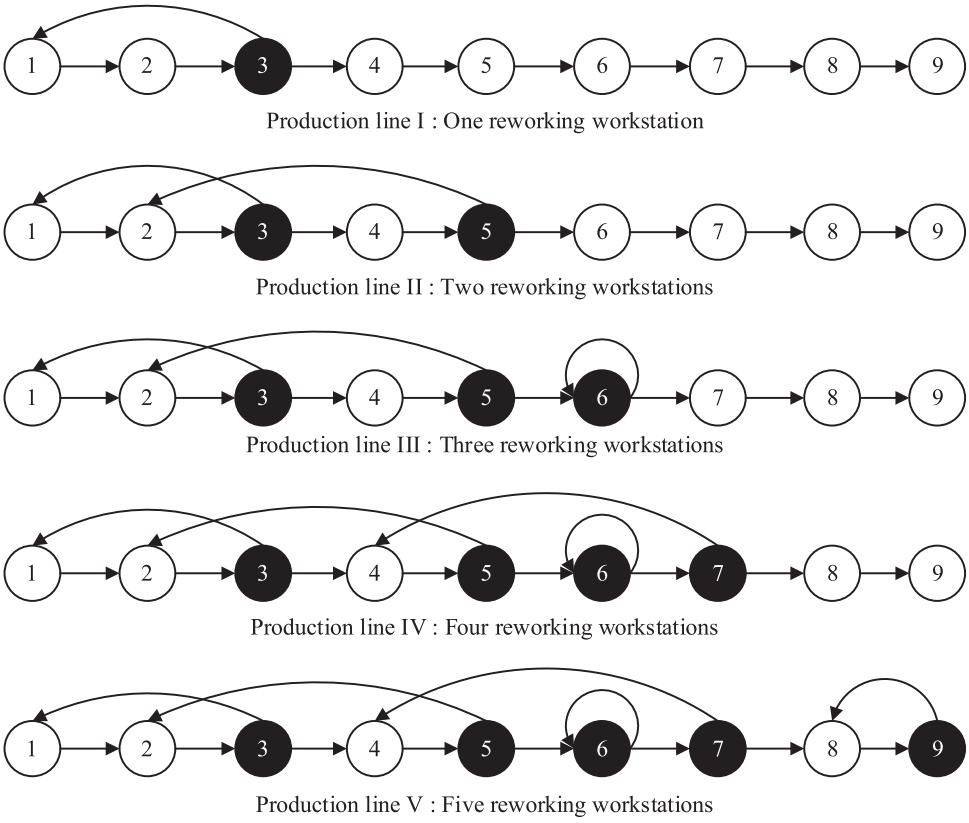


Fig. 3. Five production lines with 1 to 5 reworking stages respectively.

3.3. Comparison between the AMC-based algorithm and the graphic technique

This section assumes five production lines, all of which have identical production stages but different numbers of reworking stages (see Fig. 3), to demonstrate the computational efficiency of the AMC-based algorithm by comparing it with the graphic technique in terms of determining the required raw material input I_j and the corresponding flow vector G for each production line. For each production line, we observe each approach's central processing unit (CPU) time to determine I_j and G while maintaining the same demand and yield rates. Because the CPU time of each approach to calculate I_j and G is very short ($<$ 1 second), each approach is implemented 20,000 times for each production line to gain its total CPU time. The experimental results (see Fig. 4) show that the AMC-based algorithm has better computational efficiency in terms of determining I_j and G than that of the graphic technique as the number of reworking stages in a production line exceeds 2.

4. AMC-GA to search for the best machine configuration

The following procedure describes the AMC-GA process for integrating GA and AMC to solve the addressed problem, and its modules are illustrated in the following subsections.

Input: Demand d, HFSMS (**E**, **V**) including production lines $\{L_1, L_2, \ldots, L_m\}$, reworking stages $\{v_{j,\beta_1}, v_{j,\beta_2}, \ldots, v_{j,\beta_z}\}$ and starting stages $\{v_{j,\alpha_1}, v_{j,\alpha_2}, \ldots, v_{j,\alpha_z}\}$, number of stage-$i$ machine suppliers s_i, the attributes of the uth stage-i supplier including unit cost $c_{u,i}$, machine reliability $r_{u,i}$, production capacity $k_{u,i}$, yield rate $p_{u,i}$, and number of available machines M_i^u, B (budget), S_P (population size), R_C (crossover rate), R_M (mutation rate), R_E (elite rate), and T_{end} (terminal condition).

Output: The best machine configuration with maximum network reliability.

Step 1. Initialize the population (see Section 4.1).

Step 2. Evaluate the fitness value for each chromosome (T, Y) based on the total configuration cost $C(T, Y)$ and the network reliability $R_d(T, Y)$, where $R_d(T, Y)$ is calculated by using the AMC-based algorithm. (see Section 4.2)

Step 3. Record the best chromosome of the current generation. If the terminal condition T_{end} is fulfilled, output the best chromosome; otherwise, implement the evolution process.

Step 4. Evolution process. (see Section 4.3)

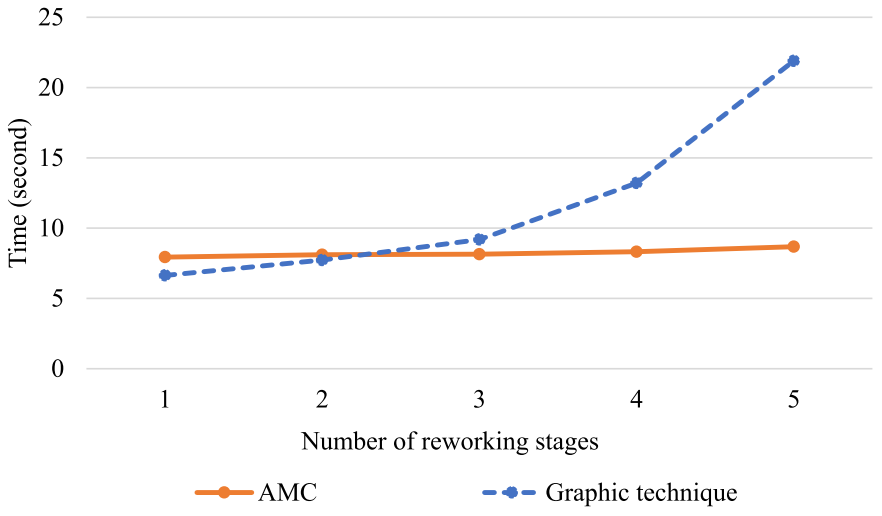


Fig. 4. Computational efficiency comparison of the AMC and the graphic technique.

Step 4.1. Parent selection.
Step 4.2. Parent crossover.
Step 4.3. Offspring mutation.
Step 4.4. Repeat the evolution process until $(1 - R_E) \times S_P$ offspring is produced.
Step 5. Execute elite replacement to create a new population. Then, go to Step 2. (see Section 4.4)

4.1. Chromosome representation and population initiation

In the AMC-GA, the notation (T, Y) of the machine configuration continues to represent a chromosome. Let S_p be the population size, which is the number of chromosomes in a population; then, Step 1 is used to generate a set of S_p chromosomes. The following equations are proposed to initialize a chromosome (T, Y)

$$t_{j,i} = Rand(1, s_i) \text{ for } i = 1, 2, \ldots, n \text{ and } j = 1, 2, \ldots, m. \tag{27}$$

$$y_{j,i} = \begin{cases} \text{Rand}\left(1, M_i^{t_{j,i}}\right) & \text{if } \ j = 1 \\ \text{Rand}\left(1, M_i^{t_{j,i}} - \sum_{e=1}^{j-1} y_{e,i}\right) & \text{if } \ j > 1 \end{cases} \text{ for } i = 1, 2, \ldots, n \text{ and } j = 1, 2, \ldots, m. \tag{28}$$

Rand(•) is a random function that selects a random integer between a given interval. Eq. (27) selects one stage-i supplier, and Eq. (28) determines the number of stage-i machines provided by the selected supplier. A chromosome can be initialized in this manner without generating the one that violates Constraints (6)–(8). Furthermore, both equations must be repeated S_P times to initialize the population.

4.2. Fitness value evaluation

The fitness value is used to measure the quality of each chromosome. Because the objective is to maximize the network reliability, $R_d(T, Y)$ is an appropriate fitness value for a chromosome. However, the chromosome may violate the budget or reliability threshold constraint (i.e., Constraint (5)) because of the random search characteristic of GA. The following two drawbacks can occur if the infeasible chromosome cannot enter the evolution process: (i) the GA cannot implement the selection operator when all chromosomes are infeasible, and (ii) the gene sequence of the infeasible chromosome is difficult to retain, allowing the phenomenon of premature convergence to occur. Therefore, penalizing the infeasible chromosome with a lower fitness value can increase the diversity of the search space [42,43]. Let $R_{\min}$ be the smallest reliability of the feasible chromosomes in the current generation; the following rules are proposed to penalize the infeasible chromosomes that may not satisfy Constraint (5).

I If an infeasible chromosome has $R_d(T, Y) = 0$ and $C(T, Y) \leq B$, its fitness value will be penalized to be $R_d(T, Y) = (R_{\min})^{\frac{B}{C(T,Y)}}$.
II If an infeasible chromosome has $R_d(T, Y)$ has $R_d(T, Y) = 0$ and $C(T, Y) > B$, its fitness value will be penalized to be $R_d(T, Y) = (R_{\min})^{\frac{C(T,Y)}{B}}$.

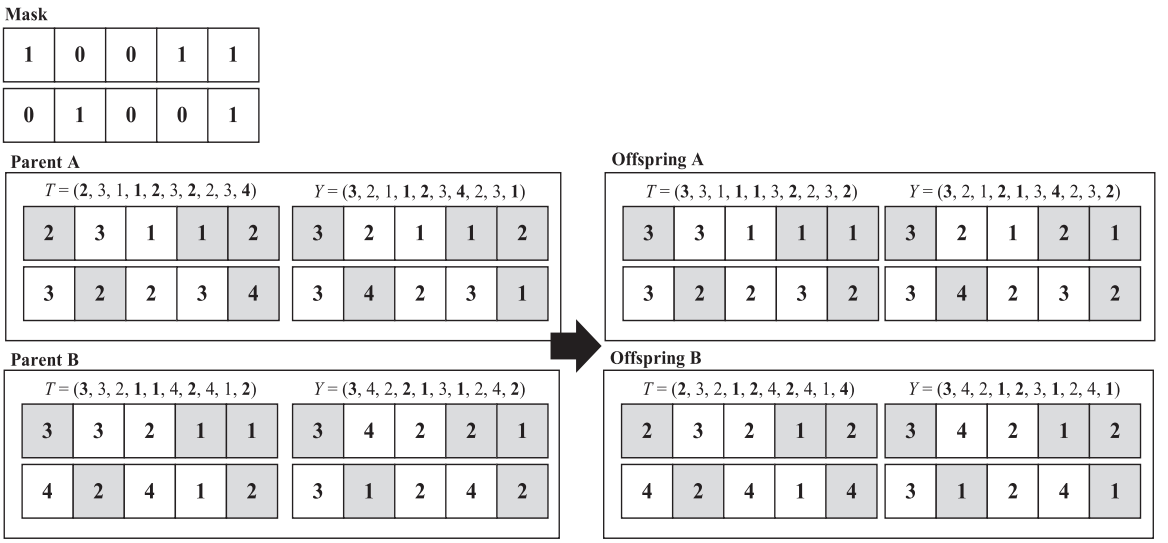


Fig. 5. Crossover representation.

III If an infeasible chromosome has $R_d(T, Y) > 0$ and $C(T, Y) > B$, its fitness value will be penalized to be $R_d = (R_{\min})^{\sqrt{\frac{C(Y)}{B}}}$.

Based on the three rules, the fitness value of the infeasible chromosome is penalized based on its cost and the lowest reliability in the current generation. The infeasible chromosomes closer to the feasible region may have higher fitness values than others but still have lower fitness values than the feasible ones. Note that if there is no feasible solution in the population, $R_{\min}$ will be given an extremely small positive value like 0.1.

4.3. Evolution process

The evolution process consists of three operators: selection, crossover, and mutation. The study uses Roulette Wheel Selection (RWS), a popular operator, to select chromosomes from the population [44]. After selecting two chromosomes as parents, the crossover operator is implemented to exchange the partial genes between the parents based on the crossover rate R_C. It is used to enhance the intensification of AMC-GA searchability. The uniform crossover is not only suitable for some equipment configuration problems [45,46] but also has better convergence [44,46]. Thus, our study uses the uniform crossover to generate a pair of offspring based on Bai, Li, Gao, and Xie's [44] study. An example of a uniform crossover is shown in Fig. 5. A $m \times n$ binary mask is randomly generated, and then the gene sequences of the two chromosomes' T and Y are reversed to produce the new offspring based on the position of the element in the matrix, which is equal to 1. If the offspring violate Constraint (6), they will be repaired by reducing the excess.

After crossover, the mutation operator implements the change in gene values of some offspring genes based on the mutation rate R_M, and it is used to enhance the diversification of AMC-GA searchability. The uniform mutation is suitable for some network reliability problems [37,47], and some equipment configuration problems [45,46]; thus, the uniform mutation based on Yeh [37] is used in our study. The uniform mutation algorithm also randomly generates an $m \times$ n binary mask and then mutates the gene sequence based on the element's position in the matrix, which is equal to 1. An example of the uniform mutation of two offspring is shown in Fig. 6. After deducting the use of other production lines, the genes that must be mutated will be randomly selected from suppliers that can supply machines, and the number of configured machines will be randomly assigned within the available quantity. Particularly, crossover and mutation operators can avoid the offspring violating Constraints (6)–(8).

4.4. Elite replacement

The elite replacement is used to reserve the elite chromosomes for the next population. Let R_E represent an elite rate. Before executing the evolution process, $R_E \times S_P$ chromosomes with fitness values in the current population are reserved. Subsequently, the evolution process is repeated $\frac{S_P(1-R_E)}{2}$ times to produce $(1 - R_E) \times S_P$ offspring. Finally, the new population is composed of the reserved chromosome and the offspring of the elite replacement.

5. Numerical analysis

In this section, three examples are adopted to show the computational performance of the proposed AMC-GA. The first example is a simple random HFSMS and is utilized to demonstrate the efficiency of the AMC-GA in searching for the exact

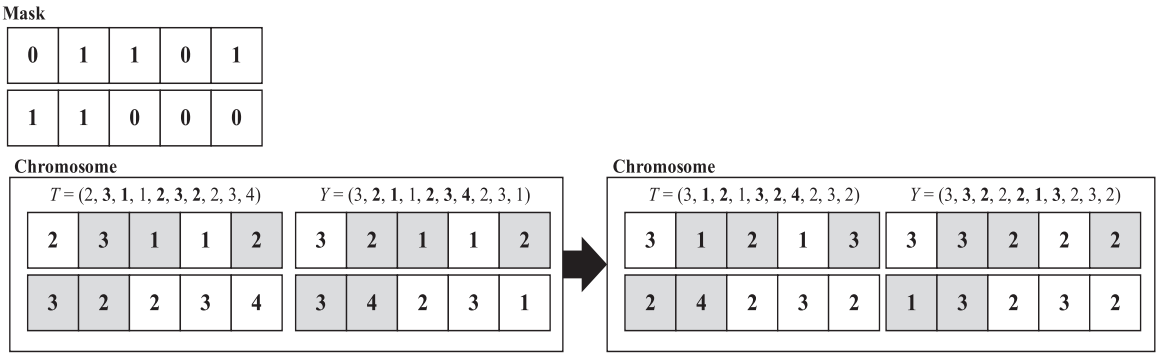


Fig. 6. Mutation representation.

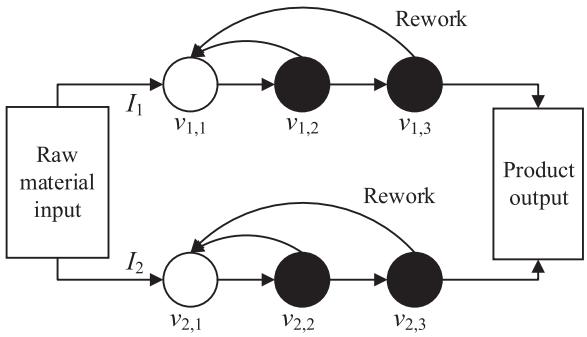


Fig. 7. A simple HFSMS with two production lines.

Table 1
The machine data of Fig. 7.

Production stage: $v_{j,i}$	Supplier: u	Capacity (pcs/per hour): $k_{u,i}$	Yield rate: $p_{u,i}$	Machine reliabilty: $r_{u,i}$	Machine cost: $c_{u,i}$	Available quantity: M_i^u
$v_{1,1}/v_{2,1}$	1	3500	0.925	0.913	150	3
	2	2800	0.905	0.962	180	4
	3	2000	0.950	0.990	200	4
	4	1600	0.988	0.934	190	5
$v_{1,2}/v_{2,2}$	1	4000	0.930	0.920	170	3
	2	3600	0.955	0.980	180	2
	3	3000	0.912	0.935	165	4
$v_{1,3}/v_{2,3}$	1	5000	0.880	0.943	130	3
	2	4500	0.905	0.960	170	2
	3	4000	0.924	0.925	150	3

or approximate solutions by comparing it with a brute-force method (BFM). The others are two practical HFSMSs related to solar cell manufacturing and printed circuit board (PCB) manufacturing, respectively, and are used to demonstrate the applicability and computational efficiency of the AMC-GA. In addition to the AMC-GA, PSO, SA, gray wolf optimizer (GWO), and teaching-learning-based optimization (TLBO), integrated with the AMC-based algorithm, namely, AMC-PSO, AMC-SA, AMC-GWO, AMC-TLBO, are also applied to the three examples, where the four meta-heuristic algorithms are programmed with references to Yeh and Fiondella [48], Du and Swamy [49], Panwar and Deep [50], and Lotfipour and Afrakhte [51], respectively. The algorithms are programmed in Python, and the experiments are executed on a personal computer with Windows 11, Intel Core i7–11375H, CPU 3.30 GHz, and 64 GB RAM.

5.1. Numerical experiments of a simple random HFSMS

The simple random HFSMS owns two production lines and three production stages, including two reworking actions in each production line (see Fig. 7). The machine data are assumed in Table 1. Two scenarios of $(d, B) = (80, 2500)$ and $(d, B) = (100, 3300)$ are considered, where demand is counted per unit of 100. For each scenario, the BFM is initially implemented to obtain the exact optimal network reliability. Subsequently, each meta-heuristic algorithm runs until the difference between the obtained maximum network reliability, and the exact value is less than or equal to 0.005.

Table 2
Experimental results of Fig. 7.

Demand (unit: 100), Budget	Meta-heuristic algorithm	Maximum reliability	Cost	Difference	CPU time	Optimal configuration
$d = 80$, B= \$2500	BFM	0.982612652526581	2460	–	1590	((1,1,2,2,2,1), (3,3,2,3,2,2))
	GA	0.982612652526581	2460	0	3.42	((1,1,2,2,2,1), (3,3,2,3,2,2))
	PSO	0.982612652526581	2460	0	19.85	((2,2,1,1,1,2), (3,2,2,3,3,2))
	SA	0.980340610371285	2460	0.002272042	20.16	((1,1,1,2,2,2), (3,3,2,3,2,2))
	GWO	0.978262135455321	2470	0.004350517	16.04	((2,1,1,1,2,2), (3,3,2,3,2,2))
	TLBO	0.979249225767988	2460	0.003363427	15.27	((1,2,1,2,1,2), (3,2,2,3,3,2))
$d = 100$, B= \$3300	BFM	0.985611751607353	3270	–	1733	((2,3,1,3,2,2), (4,4,3,4,2,2))
	GA	0.982911476905789	3030	0.002700275	6.19	((2,3,1,1,2,3), (4,4,3,3,2,3))
	PSO	0.982791790151926	3270	0.002819961	7.23	((2,3,2,3,2,1), (4,4,2,4,2,3))
	SA	0.982791790151926	3270	0.002819961	37.53	((2,3,2,3,2,1), (4,4,2,4,2,3))
	GWO	0.982791790151926	3270	0.002819961	41.86	((2,3,2,3,2,1), (4,4,2,4,2,3))
	TLBO	0.982911476905789	3030	0.002700275	51.43	((1,2,3,2,3,1), (3,2,3,4,4,3))

*Bold numbers indicate exact values.

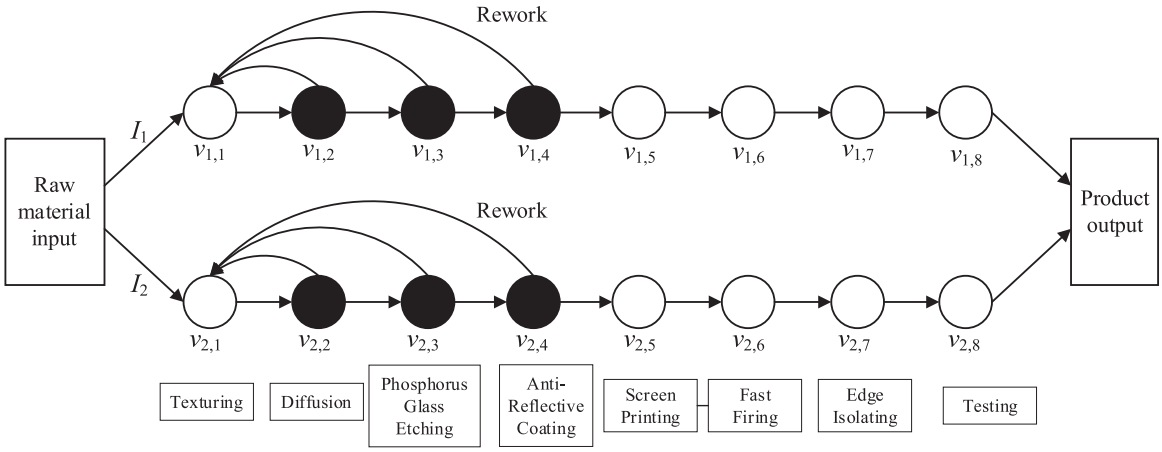


Fig. 8. The solar cell manufacturing system [52,53].

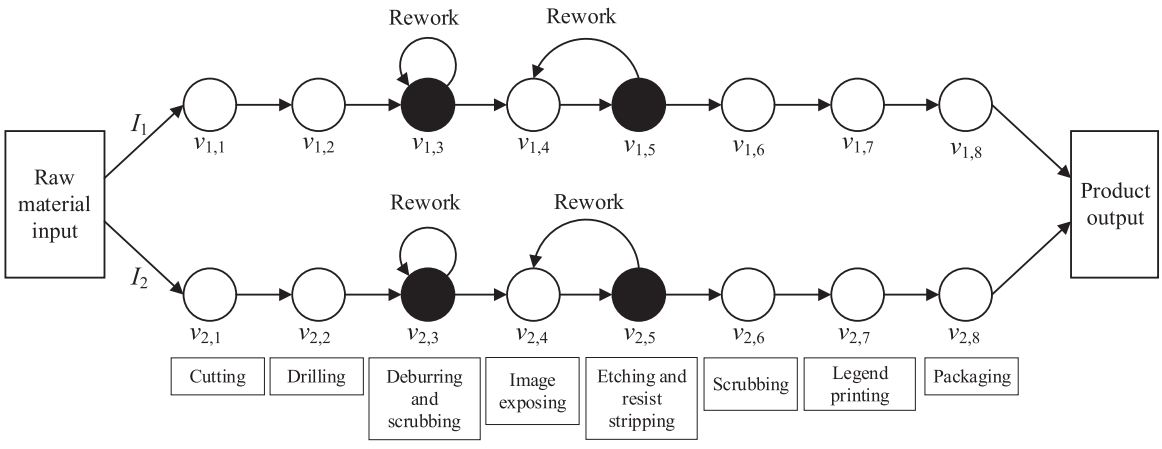


Fig. 9. The PCB manufacturing system [54].

The experimental results of both scenarios are shown in Table 2. For both scenarios, all meta-heuristic algorithms can determine high-quality solutions in a shorter time. Accordingly, the meta-heuristic algorithms can be used in larger cases. Furthermore, both AMC-GA and AMC-PSO can obtain the exact solution for the scenario of $(d, B) = (80, 2500)$ much more efficiently than the BFM and have less computational time than the other meta-heuristic algorithms for both scenarios.

Table 3
The machine data of Fig. 8.

Production stage: $v_{j,i}$	Supplier: u	Capacity (pcs/per hour): $k_{u,i}$	Yield rate: $p_{u,i}$	Machine reliabilty: $r_{u,i}$	Machine cost (ten thousand) NTD): $c_{u,i}$	Available quantity: M_i^u
$v_{1,1}/v_{2,1}$	1	2000	0.983	0.9854	122	5
	2	5000	0.978	0.9890	280	2
	3	3600	0.974	0.9872	200	4
	4	2200	0.985	0.9853	183	5
	5	2400	0.979	0.9820	186	6
$v_{1,2}/v_{2,2}$	1	1600	0.974	0.9884	153	7
	2	2000	0.978	0.9795	165	7
	3	2000	0.969	0.9910	180	5
	4	1000	0.975	0.9921	125	8
$v_{1,3}/v_{2,3}$	1	4000	0.968	0.9852	182	3
	2	3600	0.973	0.9830	176	3
	3	4000	0.976	0.9862	202	3
$v_{1,4}/v_{2,4}$	1	3600	0.977	0.9923	92	4
	2	4800	0.979	0.9894	126	3
$v_{1,5}/v_{2,5}$	1	3000	0.994	0.9915	107	3
	2	2800	0.998	0.9796	103	5
	3	1400	0.993	0.9890	81	6
	4	600	0.989	0.9920	62	10
	5	990	0.988	0.9888	65	8
$v_{1,6}/v_{2,6}$	1	5300	0.986	0.9784	144	2
	2	3000	0.988	0.9835	113	4
	3	2100	0.984	0.9822	106	4
$v_{1,7}/v_{2,7}$	1	3600	0.986	0.9654	131	3
	2	900	0.984	0.9788	75	7
	3	1500	0.98	0.9817	112	6
$v_{1,8}/v_{2,8}$	1	1500	0.998	0.9863	67	7
	2	2500	0.992	0.9830	92	5
	3	1000	0.995	0.9745	61	8
	4	3000	0.993	0.9780	106	6

Table 4
The machine data of Fig. 9.

Production stage: $v_{j,i}$	Supplier: u	Capacity (boards/day): $k_{u,i}$	Yield rate: $p_{u,i}$	Machine reliabilty: $r_{u,i}$	Machine cost (unit: 1000 NTD): $c_{u,i}$	Available quantity: M_i^u
$v_{1,1}/v_{2,1}$	1	4500	0.985	0.987	252	5
	2	5000	0.956	0.946	260	6
	3	4000	0.937	0.916	195	4
$v_{1,2}/v_{2,2}$	1	4000	0.963	0.996	285	6
	2	4600	0.947	0.956	316	5
	3	5200	0.977	0.988	427	4
	4	3200	0.924	0.924	205	6
$v_{1,3}/v_{2,3}$	1	3500	0.960	0.985	368	7
	2	4000	0.983	0.926	363	5
	3	3200	0.914	0.906	220	5
$v_{1,4}/v_{2,4}$	1	1500	0.987	0.964	198	10
	2	1200	0.934	0.913	115	8
$v_{1,5}/v_{2,5}$	1	3000	0.974	0.975	346	8
	2	3600	0.988	0.964	450	4
	3	2200	0.902	0.915	230	5
$v_{1,6}/v_{2,6}$	1	3500	0.970	0.984	324	6
	2	3800	0.973	0.977	396	4
	3	2800	0.985	0.956	424	6
	4	3300	0.917	0.914	236	3
	5	2600	0.922	0.907	240	5
$v_{1,7}/v_{2,7}$	1	4500	0.992	0.996	176	4
	2	2000	0.990	0.998	195	8
$v_{1,8}/v_{2,8}$	1	4000	0.995	0.998	341	6
	2	4400	0.996	0.999	390	6
	3	3800	0.982	0.932	199	4

Table 5
Comparison among meta-heuristic algorithms for Fig. 8.

Demand (unit:100)	Meta-heuristic algorithm	Average maxmum network reliability	Largest maximal network reliability	Corresponding configuration cost (unit:10,000 NTD)[b]	Best solution: (T, Y)
$d = 60$	GA	0.993674479[a]	0.996079166[a]	4997	((1,3,1,1,2,1,1,2,3,2,2,1,1,2,3,1), (3,3,2,2,3,2,2,3,2,3,2,2,2,2,3,3))
	PSO	0.987600435	0.992261269	4973	((3,3,1,1,1,2,1,1,1,2,3,1,2,1,3,1), (2,2,2,2,2,2,2,2,3,3,3,2,2,3,2,4,4))
	SA	0.967764731	0.977269869	4985	((1,2,1,2,2,2,1,4,3,1,3,1,2,1,2,1), (3,2,2,1,2,2,3,2,2,3,2,2,3,2,6,4))
	GWO	0.97082463	0.985955615	4994	((3,3,1,1,3,3,3,2,1,3,3,2,1,2,1,2), (1,1,1,1,2,1,2,1,5,4,3,3,3,4,3,3))
	LTBO	0.984057206	0.990046446	4971	((1,2,3,1,2,2,1,1,1,3,1,1,1,1,2,2), (2,3,2,2,2,2,2,4,3,3,2,2,2,2,6,3)
$d = 70$	GA	0.970716703[a]	0.985573305[a]	4996	((1,2,1,1,2,2,1,2,3,3,2,1,1,2,2,1), (5,5,3,3,4,3,3,4,1,1,1,1,1,3,2))
	PSO	0.956252965	0.982102398	4967	((3,3,1,1,1,2,1,1,4,2,3,1,2,2,2,2), (3,4,3,3,3,3,3,3,6,1,1,1,1,2,1,3,2))
	SA	0.929527642	0.952804867	4952	((3,1,1,1,1,3,2,1,1,3,3,2,2,1,1,2), (1,2,1,1,1,2,5,2,5,4,2,2,3,2,3,4))
	GWO	0.916673655	0.935282394	5000	((2,2,1,2,1,1,1,2,2,3,1,1,3,3,3,1), (1,3,2,2,2,2,2,3,1,3,1,3,4,3,4,3))
	LTBO	0.931443351	0.947983605	4983	((2,1,3,1,1,2,1,4,2,1,2,2,2,1,2,2), (1,3,1,2,2,2,2,2,1,4,3,1,3,2,7,3))
$d = 80$	GA	0.90269455[a]	0.930398417[a]	4999	((1,2,2,1,2,2,1,2,3,3,3,1,1,3,3,1), (4,4,3,3,3,3,3,4,1,2,1,1,2,2,2,2))
	PSO	0.849613268	0.886038166	4988	((3,2,3,1,1,1,1,2,2,2,3,1,2,2,2,1), (2,4,2,3,2,2,3,3,1,3,1,1,2,2,5,3))
	SA	0.847508607	0.888749216	4997	((2,3,1,1,1,1,1,1,3,1,2,1,3,3,3,1), (2,5,3,3,3,2,3,6,1,1,1,1,1,3,2,1))
	GWO	0.828348213	0.860055634	4892	((3,3,3,1,1,2,1,4,3,2,1,1,1,3,2,2), (1,2,1,1,1,2,2,2,2,4,2,3,2,4,7,4))
	LTBO	0.861622085	0.895897242	5000	((1,3,3,1,2,2,3,4,3,1,3,1,1,2,1,1), (4,4,2,2,4,3,5,3,1,2,1,2,1,1,2,3))

[a] Bold numbers represent the best values for the column in the corresponding scenario.
[b] The cost corresponds to the largest maximal network reliability.

Table 6
Comparison among meta-heuristic algorithms for Fig. 9.

Budget (unit:1000)	Meta-heuristic algorithm	Average maximal network reliability	Largest maximal network reliability	Corresponding configuration cost (unit:1000 NTD)[b]	Best solution: (T, Y)
$B = 9400$	GA	0.909380399[a]	0.945334295[a]	9374	((1,1,3,1,1,1,1,3,1,1,1,1,1,1,1,2), (3,3,4,6,3,3,2,4,1,1,1,2,1,1,1,1))
	PSO	0.838289167	0.935193210	9378	((3,1,3,1,1,1,1,3,1,1,1,2,1,1,1,2), (4,3,4,6,3,3,2,3,1,1,1,5,1,1,1,1))
	SA	0.818372479	0.860737214	9351	((3,1,1,1,1,4,1,1,2,1,1,2,2,2,1,2), (4,2,2,5,3,3,2,2,2,1,1,4,2,1,1,1))
	GWO	0.780058834	0.820227747	9301	((3,1,3,2,1,2,1,1,1,1,1,4,3,2,1,3), (3,2,3,7,3,2,2,2,1,1,1,4,5,1,1,2))
	LTBO	0.867858329	0.933902238	9304	((1,1,3,1,1,1,1,2,1,1,1,1,1,1,1,2), (2,3,4,6,3,3,2,2,1,1,1,3,1,1,1,1))
$B = 10{,}200$	GA	0.970773703[a]	0.982763355[a]	10,197	((1,1,3,1,1,1,1,2,1,1,1,2,1,1,1,1), (3,3,5,7,3,3,3,2,1,1,1,6,1,1,1,1))
	PSO	0.944806903	0.979281364	10,162	((1,1,1,1,1,1,1,2,1,1,1,1,1,1,1,3), (3,3,3,7,3,3,3,2,1,1,1,3,1,1,1,2))
	SA	0.874446241[a]	0.896076716[a]	10,024	((1,3,1,1,1,2,1,2,1,4,3,2,1,4,1,3), (2,2,3,7,3,3,2,2,1,2,1,3,1,2,1,3))
	GWO	0.876383275	0.914916875	10,163	((1,1,3,2,1,1,1,2,1,1,1,1,1,1,2,1,2), (3,2,4,8,4,3,3,2,1,1,2,4,1,1,1,1))
	LTBO	0.93052700	0.974833654	10,166	((1,1,1,1,1,1,1,1,2,3,1,3,1,1,1,1,1,2), (3,3,3,7,3,3,3,2,2,1,2,2,1,1,1,1))
$B = 11{,}000$	GA	0.987362098[a]	0.993776261[a]	10,971	((1,1,3,1,1,1,1,2,1,1,1,2,1,1,1,2), (3,4,5,7,4,4,3,2,1,1,1,4,1,1,1,1))
	PSO	0.974760487	0.990984582	10,900	((3,1,3,2,1,1,2,1,1,1,1,1,1,1,1,2), (2,1,2,4,1,1,1,1,3,3,3,7,4,4,3,2))
	SA	0.920584847	0.962204959	10,676	((2,1,3,1,1,1,1,2,1,2,1,1,2,3,2,3), (3,3,4,7,4,3,3,2,1,2,1,2,1,1,2,1))
	GWO	0.908789902	0.966072822	10,857	((1,4,1,1,1,2,1,2,1,1,3,1,2,2,1,2), (3,4,3,7,3,3,3,2,2,1,3,2,1,1,1,1))
	LTBO	0.953820411	0.989457601	10,862	((1,1,3,1,1,1,1,2,1,1,1,1,3,1,1,2), (4,3,5,7,4,3,3,2,1,1,1,3,2,1,1,1,,))

[a] Bold numbers represent the best values for the column in the corresponding scenario.
[b] The cost corresponds to the largest maximal network reliability.

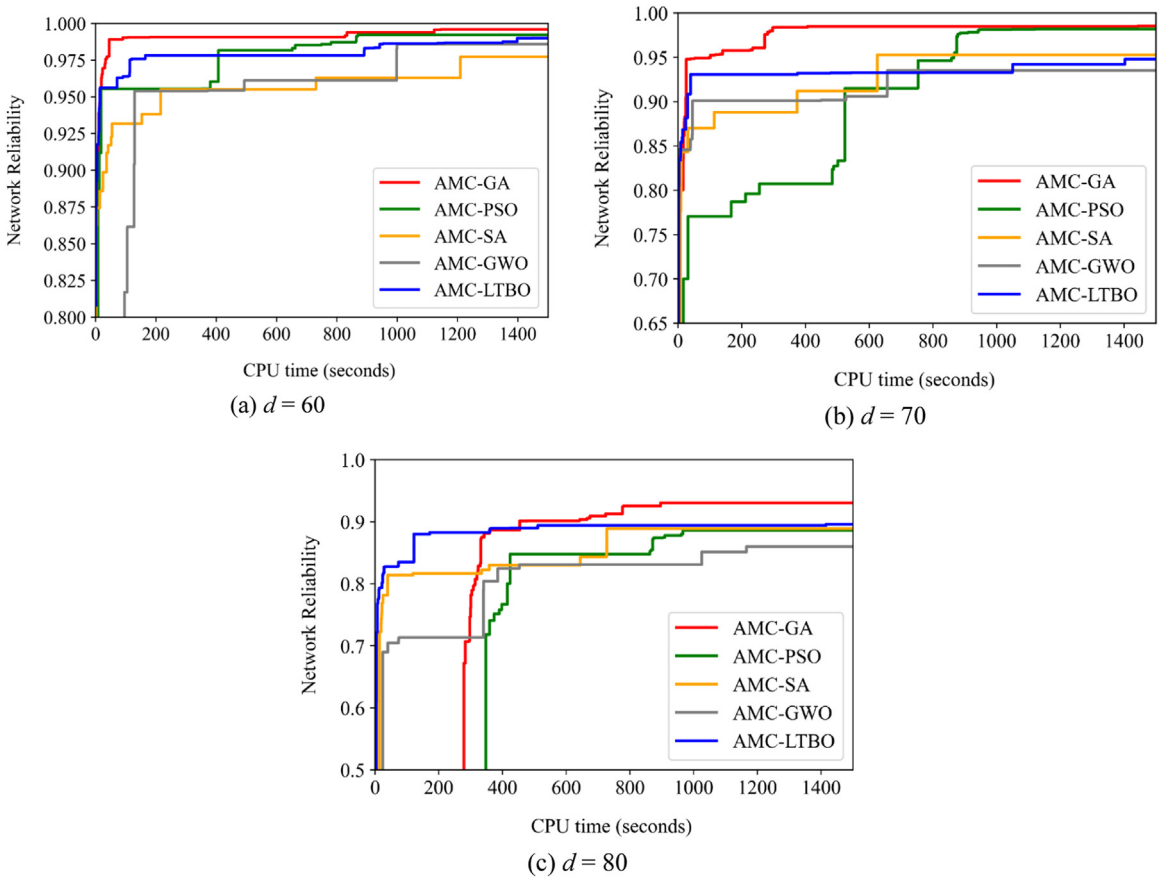


Fig. 10. Comparison among the meta-heuristic algorithms for Fig. 8.

5.2. Numerical experiments of solar cell manufacturing and PCB manufacturing

The first practical case is the solar cell manufacturing system. By referring to Song and Lin [52] and Chen, Chen, and Chou [53], each solar cell production line must contain eight stages to convert wafers into solar cells: (1) Texturing, (2) Diffusion, (3) Phosphorus Glass Etching, (4) Anti-Reflective Coating, (5) Screen Printing, (6) Fast Firing, (7) Edge Isolating, and (8) Testing. At each production stage, several parallel machines perform the same process. Additionally, each production line has three reworking stages, including diffusion, phosphorus, and anti-reflective coating production stages. These reworking stages convey defective products to the texturing stage. For instance, Fig. 8 shows the solar cell manufacturing system with two production lines.

The second practical case is the PCB manufacturing system. By referring to Yeh, Chang, and Chen [54], each PCB production line includes eight stages which are (1) Cutting, (2) Drilling, (3) Deburring-scrubbing, (4) Image-exposing, (5) Etching-resist-stripping, (6) Scrubbing, (7) Legend printing, and (8) Packaging. Similar to solar cell manufacturing, several identical machines are configured in parallel to implement the same process. Each production line owns both reworking stages of deburring-scrubbing and etching-resist-stripping. The defective flow from the deburring-scrubbing stage will be repaired at the same stage, and the defective flow from the etching-resist-stripping stage will be sent to the image-exposing stage for repair. Fig. 9 depicts this PCB manufacturing system. Tables 3 and 4 summarize the technical specifications of the machines depicted in Figs. 8 and 9. The data can be gathered from various sources, including historical usage experience and reliable databases provided by the companies and suppliers, and give an understanding of each machine's capability, reliability, yield rate, and purchase cost.

For the solar cell case, three scenarios are set as $(d, B) = (60, 5000)$, $(d, B) = (70, 5000)$, and $(d, B) = (80, 5000)$ for observing the influence of demand on the network reliability. For the PCB case, three scenarios of $(d, B) = (80, 9400)$, $(d, B) = (80, 10{,}200)$, and $(d, B) = (80, 11{,}000)$ are set for observing how the budget affects the network reliability. Because finding the exact optimal solution by the BFM is time-consuming or may occur an out-of-memory error, we focus on comparing the AMC-GA with the other meta-heuristic algorithms. The parameter settings affect AMC-GA convergence, particularly for the population size S_P, crossover rate R_C, mutation rate R_M, and elite rate R_E. By the recommendations from several references [55–58] and our pretest, the AMC-GA parameter setting is given with $(S_P, R_C, R_M, R_E) = (100, 0.6, 0.025, 0.1)$. For each

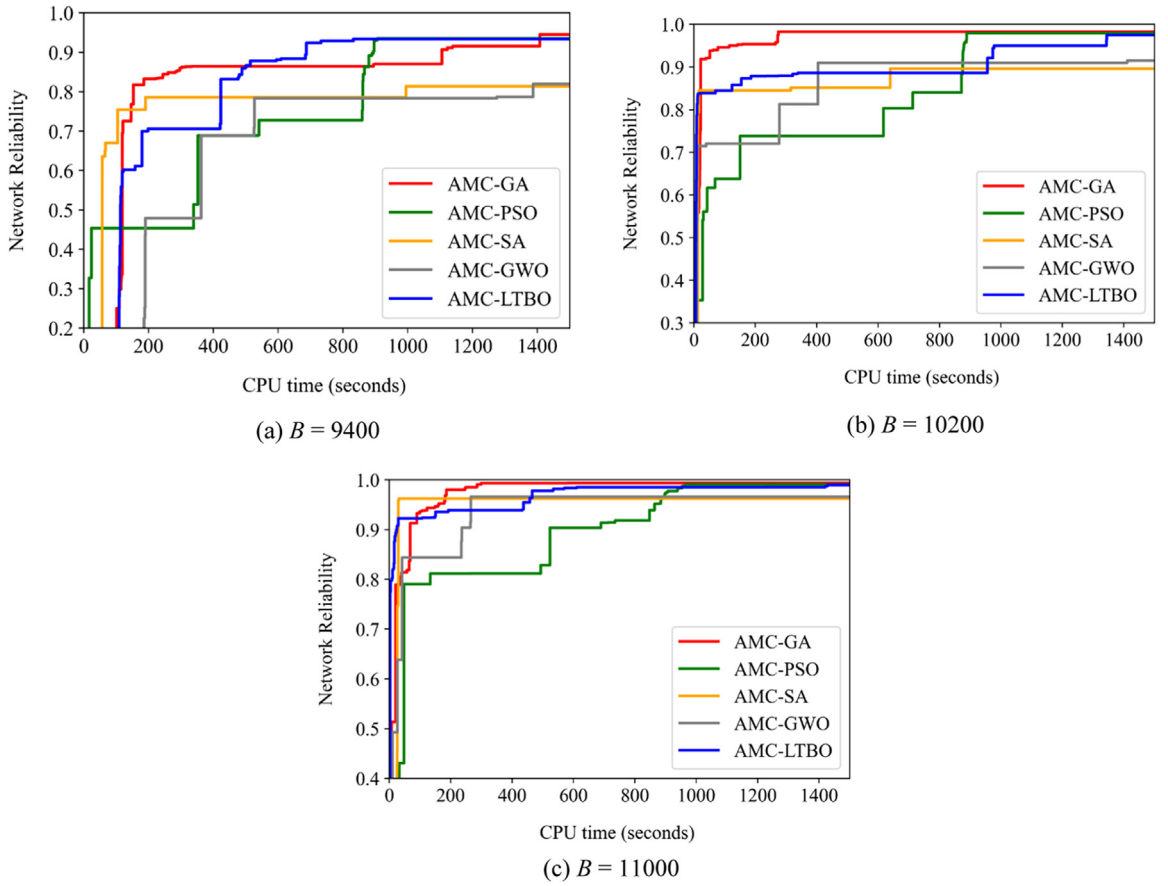


Fig. 11. Comparison among the meta-heuristic algorithms for Fig. 9.

scenario, each algorithm is implemented for ten trials to determine the average maximum network reliability and the largest maximum network reliability, as well as the corresponding configuration cost and best machine configuration. Additionally, each algorithm runs for 1500s (i.e., $T_{end} = 1500$s) in each trial.

Table 5 summarizes the comparison results of Fig. 8. In Table 5, the AMC-GA performs best regarding the largest value and average value for the three scenarios in the solar cell case. The other algorithms perform worse, especially for the scenario $(d, B) = (80, 5000)$. The experiment also provides a decision reference for a system administrator. According to the AMC-GA results, the system administrator can use the configuration $(T, Y) = ((1, 2, 1, 1, 2, 2, 1, 2, 3, 3, 2, 1, 1, 2, 2, 1), (5, 5, 3, 3, 4, 3, 3, 4, 1, 1, 1, 1, 1, 1, 3, 2))$ to design the system such that it has more than 0.99 probability to satisfy the demand $d = 70$ (unit: 100) under the budget $B = 5000$ (unit: 10,000 NTD), where $t_{1,1} = 1$ indicates that the 1st supplier of production stage $v_{1,1}$ is selected and $y_{1,1} = 5$ indicates that the number of configured machines is 5. However, if the system administrator expects that the system can fulfill the demand $d = 80$ with the network reliability ≥ 0.99, he/she should increase the budget or collaborate with other appropriate suppliers. Fig. 10 illustrates the optimization process of the best trial of each meta-heuristic algorithm under various demand scenarios. The AMC-GA displays better convergence than the other algorithms. In particular, Fig. 10(c) shows that the AMC-GA can quickly search for a better solution once it gets a feasible solution.

The comparison results of Fig. 9 are given in Table 6 and display that the AMC-GA also performs best regarding the largest value and average value for the three scenarios in the solar cell case. The followings are the AMC-PSO and AMC-TLBO. The AMC-SA and AMC-GWO perform worst. As the budget increases, the average network reliability of each algorithm also increases. However, the AMC-GA can still get average values that exceed 0.9 and has a slight difference between the average and the largest values. That is, the AMC-GA shows excellent stability. The experimental results also provide a decision reference. For instance, if the system administrator expects that the system has more than 0.99 network reliability with $d = 80$, he/she can consider the alternative $(T, Y) = ((1, 1, 3, 1, 1, 1, 1, 2, 1, 1, 1, 2, 1, 1, 1, 2), (3, 4, 5, 7, 4, 4, 3, 2, 1, 1, 1, 4, 1, 1, 1, 1))$ to design the system. However, the system administrator should consider demand reduction or collaborate with other appropriate suppliers if the budget is insufficient. Fig. 11 illustrates the optimization process of the best trial of each meta-heuristic algorithm under various demand scenarios. Although the solutions obtained by the AMC-GA may be

worse than the other algorithms, the AMC-GA can terminate with the best solution in each scenario. Furthermore, Fig. 11(a) presents that the AMC-GA still has the chance to find a better solution if more terminal time can be given.

6. Conclusions

Designing a reliable HFSMS with reworking and scraping actions is critical for a system administrator. Our study focuses on determining the optimal machine configuration with maximum network reliability of the HFSMS using the AMC-GA, where the AMC-based algorithm is proposed to calculate the network reliability, and the GA is used to search for the optimal machine configuration. The AMC-based algorithm calculates the raw material input and the flow vector for each production line according to the transition probability matrix generated based on the yield rates of the configured machines without enumerating all regular and reworking paths as the graphic techniques of Lin and Chang [19,20], Hao et al. [21] and Zhang et al. [23]. The experimental results of five production lines, each with a different number of reworking stages, also show that the AMC-based algorithm has better computational efficiency than the graphic technique. The experiments of the simple HFSMS demonstrate that the meta-heuristic algorithms can find out the exact or approximate optimal solution more efficiently than the BFM. The solar cell and PCB HFSMSs are implemented to demonstrate the AMC-GA's computational efficiency compared with AMC-PSO, AMC-SA, AMC-GWO, and AMC-TLBO. Furthermore, the best machine configuration produced by the AMC-GA can be viewed as a decision reference for the system administrator when constructing the solar cell manufacturing system.

In some HFSMSs, the product flow may be reworked multiple times at another production line. However, the proposed AMC-based algorithm only applies when the product flow can only be reworked once at the same stage. Future research may extend the addressed problem by allowing the product flow to be reworked multiple times at other production lines and then discussing how the situation affects the HFSMS network reliability.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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